



**TRIBHUVAN UNIVERSITY  
INSTITUTE OF ENGINEERING  
PURWANCHAL CAMPUS**

**A Minor Project Report On Harnessing Satellite Imagery and IoT for  
Real - Time Landslide and Accident Risk Management**

**BY**

**Sushant Niraula (PUR078BEI047)**

**Nabin Yadav (PUR078BEI020)**

**Lokesh Kumar Mandal (PUR078BEI017)**

**Ashok Kumar Ghalan (PUR078BEI07)**

**DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING  
PURWANCHAL CAMPUS  
DHARAN, NEPAL**

**MARCH, 2025**

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## **DECLARATION**

We hereby declare that the project report entitled “**Harnessing Satellite Imagery and IoT for Real-Time Landslide and Accident Risk Management**” submitted to the Department of Electronics and Computer Engineering, IOE, Purwanchal Campus, in partial fulfillment of the requirements for the Degree of Bachelor of Engineering in Electronics, Communication, and Information Technology, is the result of our dedicated efforts and hard work.

We confirm that the materials presented in this report have not been submitted to any other university or institution for the award of any degree. Furthermore, we affirm that we are the sole authors of this work, and no sources other than those explicitly referenced have been utilized in its preparation.

**Sushant Niraula** PUR078BEI047

**Nabin Yadav** PUR078BEI020

**Lokesh Kumar Mandal** PUR078BEI017

**Ashok Kumar Ghalan** PUR078BEI07

# CERTIFICATE OF APPROVAL

The undersigned certify that they have read and recommended to the Institute of Engineering for acceptance, a project report entitled “**Harnessing Satellite Imagery and IoT for Real-Time Landslide and Accident Risk Management**” submitted by Sushant Niraula, Nabin Yadav, Lokesh Kumar Mandal, Ashok Kumar Ghalan in partial fulfillment of the requirements for the Bachelor’s Degree in Electronics and Communication Engineering.

---

Deputy Head of Department: Asst. Prof. Rajnish Rajbahak  
Department of Electronics and Computer Engineering  
Institute of Engineering, Purwanchal Campus  
Date of Approval: March, 2025

---

Supervisor of Project: Asst. Prof. Pukar Karki  
Department of Electronics and Computer Engineering  
Institute of Engineering, Purwanchal Campus  
Date of Approval: March, 2025

## ACKNOWLEDGEMENT

We express our deepest gratitude to all the individuals and institutions whose unwavering support and encouragement have made this project, “**Harnessing Satellite Imagery and IoT for Real-Time Landslide and Accident Risk Management,**” possible.

This project is dedicated to the memory of those who lost their lives due to landslides and road accidents in Nepal. Tragic incidents such as the landslide on July 12, 2024, that swept two buses carrying at least 60 people into a river, and the devastating floods and landslides on September 27 and 28, 2024, which claimed over 170 lives, serve as stark reminders of the urgent need for solutions to mitigate these risks. Countless similar incidents have resulted in the loss of thousands of innocent lives, and it is our sincere hope that this project will contribute to saving lives and reducing such tragedies in the future.

We extend our heartfelt gratitude to our parents, whose unwavering support, sacrifices, and encouragement have shaped us into the individuals we are today.

Special thanks go to our teachers for their invaluable guidance and motivation throughout our academic journey, and to our families for their constant encouragement. We are also deeply thankful to our community and society for fostering an environment that inspires innovation and service to the nation. Also we are profoundly thankful to our supervisor, **Mr Pukar Karki Sir** whose expert guidance, constructive feedback, and continuous encouragement have been invaluable throughout this project. His insights and support have greatly enhanced the quality of our work and motivated us to push our boundaries.

**Sushant Niraula** PUR078BEI047

**Nabin Yadav** PUR078BEI020

**Lokesh Kumar Mandal** PUR078BEI017

**Ashok Kumar Ghalan** PUR078BEI07

## ABSTRACT

The project “**Harnessing Satellite Imagery and IoT for Real-Time Landslide and Accident Risk Management**” aims to develop an integrated framework for landslide susceptibility mapping and real-time hazard prediction. The project is executed in two phases. In the first phase, a *Landslide Susceptibility Map (LSM)* is created using satellite-derived band data, geological surveys, and other environmental datasets to identify high-risk zones. By analyzing spectral features, terrain attributes, and past landslide occurrences, the system provides a data-driven approach for assessing susceptibility.

In the second phase, an IoT-based early warning system is developed by deploying sensors in landslide-prone areas to collect real-time data on ground movement, moisture levels, vibration, and weather conditions. These sensor readings are continuously monitored and analyzed using machine learning algorithms to assess the likelihood of landslides, enabling timely alerts and risk mitigation strategies.

The integration of satellite data and real-time IoT monitoring enhances prediction accuracy and reduces alert delays, contributing to effective and efficient disaster management. The proposed system supports proactive risk assessment and real-time hazard mitigation, making it valuable for infrastructure planning, early warning systems, and disaster response efforts. Future improvements will focus on enhancing data fusion techniques and ensuring the system’s scalability and interoperability with existing emergency management frameworks.

### **Keywords:**

1. Satellite Imagery
2. IoT (Internet of Things)
3. Landslide Susceptibility Map (LSM)
4. Early Warning System
5. Machine Learning
6. Environmental Datasets

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## **LIST OF ABBREVIATIONS**

API : Application Programming Interface

UI : User Interface

IoT : Internet of Things

GIS : Geographic Information System

ML : Machine Learning

# CHAPTER 1

## INTRODUCTION

### 1.1 Background

Nepal's fragile terrain makes it highly prone to landslides, especially along hilly roads, causing frequent disruptions and fatalities, particularly during the monsoon season. Critical routes like the Mugling-Narayanghat Highway face recurrent landslides, impacting transportation and livelihoods. Major disasters, such as the July 12, 2024, landslide that swept away two buses and the September 27-28, 2024, floods and landslides that claimed over 170 lives, highlight the urgent need for an effective landslide monitoring and early warning [1].

Despite frequent disasters, Nepal lacks real-time monitoring and landslide susceptibility mapping, leaving communities vulnerable. For a resource-limited country, cost-effective solutions are crucial for disaster mitigation. This project integrates IoT sensors, satellite-derived band data, and machine learning to:

- Map landslide-prone areas using satellite data and geological surveys.
- Deploy IoT-based early warning systems to monitor real-time conditions.
- Issue timely alerts for proactive disaster management and road safety.

By leveraging affordable and scalable technology, this project offers a sustainable, low-cost solution for disaster risk reduction, strengthening Nepal's resilience against landslides.

### 1.2 Gap Identification

Landslides and road accidents are significant threats in Nepal, particularly in hilly and mountainous regions. While various efforts have been made to monitor and mitigate these risks, existing systems have several limitations:

### 1. Lack of Real-Time Monitoring and Early Warning Systems:

- Traditional methods rely heavily on past disaster records and geological surveys, which are static and fail to provide real-time risk assessment.
- The absence of an integrated IoT-based system limits the ability to issue timely alerts to prevent casualties.

### 2. Limited Use of Satellite and GIS Data for Predictive Analysis:

- Current approaches do not fully utilize satellite imagery and GIS techniques to analyze terrain vulnerability and changes over time.
- The integration of machine learning models with satellite data for proactive disaster risk mapping is still underdeveloped and underestimated in Nepal.

### **Addressing the Gaps**

This project aims to bridge these gaps by:

- Developing an IoT-based real-time monitoring system with sensor networks in high-risk areas.
- Utilizing satellite imagery and GIS data to create a predictive machine learning model for landslide and accident risk mapping.

## **1.3 Motivation**

Frequent loss of lives and property due to landslides and landslide-related road accidents is a major concern in Nepal. We are motivated by the urge to save lives, make road travel safer, and make travel in Nepal safer and more reliable. We want to bring modern technologies such as IoT and satellite imagery together to provide real-time monitoring and early warnings that can prevent such tragedies. This project is, therefore, a step ahead in the creation of a safer, resilient Nepal and contributes toward the technological advancement of disaster management. This is just a small step towards the safer travel in Nepal. Projects and works like these in this field of Disaster Management would help Nepal and Government take proactive measures ensuring safety of life's of Thousands of people.

## 1.4 Objectives

### 1. Landslide Susceptibility Mapping:

Identify and map landslide-prone areas using satellite-derived band data, geological surveys, and machine learning models to achieve high-accuracy susceptibility prediction while ensuring economic feasibility for large-scale implementation.

### 2. IoT-Based Real-Time Monitoring and Early Warning System:

Deploy IoT sensors in high-risk areas to continuously monitor critical parameters (e.g., ground movement, moisture levels, and environmental changes). Establish an automated early-warning system to promptly alert local communities and authorities, enhancing safety and disaster response.

## 1.5 Scope of Project

This project aims to develop a real-time risk surveillance system that seamlessly integrates satellite-derived environmental data with ground-based IoT sensor networks to enhance landslide and accident risk assessment. The system will leverage multi-spectral satellite datasets from sources such as USGS/SRTMGL1-003, UCSB-CHG/CHIRPS/DAILY, MODIS/061/MOD13Q1, and ESA/WorldCover/v200/2021, capturing crucial geospatial and climatic parameters like elevation, rainfall, vegetation health, and land cover dynamics. Complementing this, a IoT sensor—including soil moisture sensor, temperature and humidity sensor, and accelerometer—will be strategically deployed in high-risk zone to provide localized, real-time environmental data for immediate risk assessment.

To generate an accurate and dynamic Landslide Susceptibility Map (LSM), these different data streams will be fused using machine learning algorithms, enhancing predictive accuracy and facilitating proactive disaster management strategies. The processed risk probabilities will be integrated into a GIS-based decision-support platform, enabling visual risk assessment(Q-GIS), spatial analysis, and informed decision-making for disaster preparedness.

Beyond susceptibility mapping, the deployed IoT system will function as an early warn-

ing mechanism, continuously monitoring real-time risk factors and issuing automated alerts to emergency responders, local communities, and transportation authorities. The system will undergo field testing and prototyping to validate its scalability, interoperability, and real-world responsiveness, ensuring reliable continuous monitoring and laying the groundwork for future advancements in disaster risk management. It would require lot of testing and real world data to function well thus making the process lengthy but would facilitate the risk management in a humongous way.

By combining geospatial intelligence with IoT-driven real-time monitoring, this integrated framework has the potential to significantly enhance proactive hazard detection, risk estimation, and real time mitigation actions, thereby strengthening disaster resilience, improving road safety, and minimizing human and economic losses in landslide-prone regions.

## CHAPTER 2

### RELATED THEORY

#### 2.1 Remote Sensing

Remote Sensing is the technique of collecting information about the Earth's surface from a distance using satellites, drones, or aircraft. It works by detecting reflected or emitted radiation to analyze land, water, and atmospheric conditions.

Applications:

1. Disaster Management: Detecting landslides, floods, and earthquakes.
2. Agriculture and Forestry: Monitoring crops and deforestation.
3. Urban Planning: Mapping land use changes.
4. Climate Studies: Tracking pollution and global warming.

Remote sensing is a vital tool for environmental monitoring, disaster prevention, and scientific research.

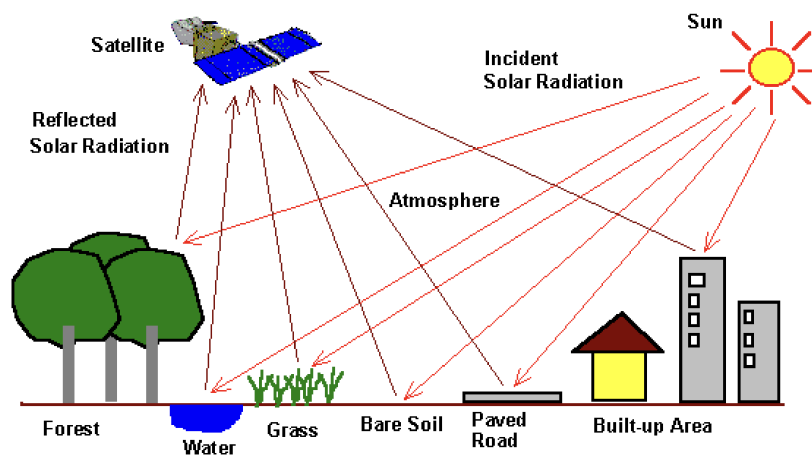


Figure 2.1: Simple demonstration of how satellite based Remote Sensing Work.

#### 2.2 GIS (Geographic Information System)

GIS (Geographic Information System) is a technology that captures, stores, analyzes, and visualizes spatial data to understand geographical patterns and relationships. It integrates maps, satellite imagery, and data layers to support decision-making in various



fields, including disaster management, urban planning, and environmental monitoring. In our project, GIS plays a crucial role in mapping landslide-prone areas, analyzing terrain changes, and integrating satellite and sensor data for risk assessment and early warning systems.

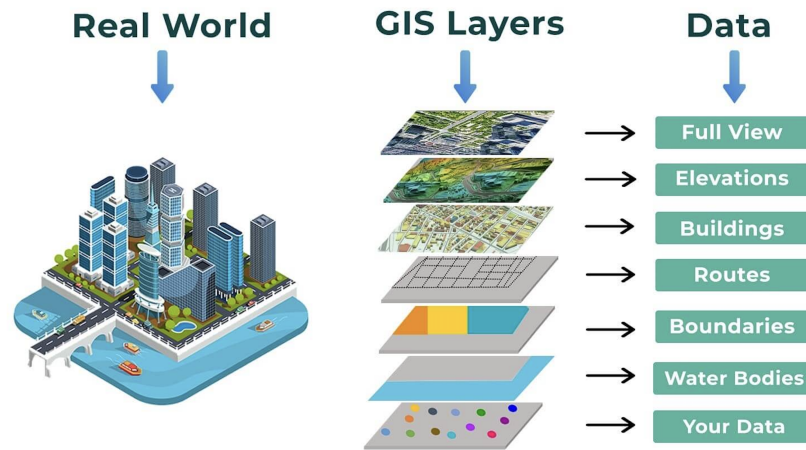


Figure 2.2: illustration of how GIS work.

### 2.3 IoT (Internet of Things)

IoT (Internet of Things) is a network of interconnected devices that collect, transmit, and analyze data in real time. These devices, equipped with sensors and microcontrollers, communicate via the internet or wireless protocols like Wi-Fi, LoRa, and MQTT. In our project, IoT plays a vital role in monitoring environmental conditions using sensors (e.g., tilt, soil moisture, and accelerometers) to detect landslides and road risks. The collected data is processed and sent to cloud platforms for analysis, enabling real-time alerts and early warning systems to enhance disaster management and road safety.

### 2.4 ESP32 and MicroControllers

ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth, widely used in IoT applications for real-time data collection and wireless communication. Microcontrollers are small computing devices that control sensors and actuators in embedded systems. In our project, ESP32 plays a key role in gathering environmental data from sensors (e.g., soil moisture, tilt, and accelerometers) and transmitting it to cloud

platforms. Its low power consumption and wireless capabilities make it ideal for remote monitoring in landslide and accident-prone areas, enabling real-time alerts and decision-making.

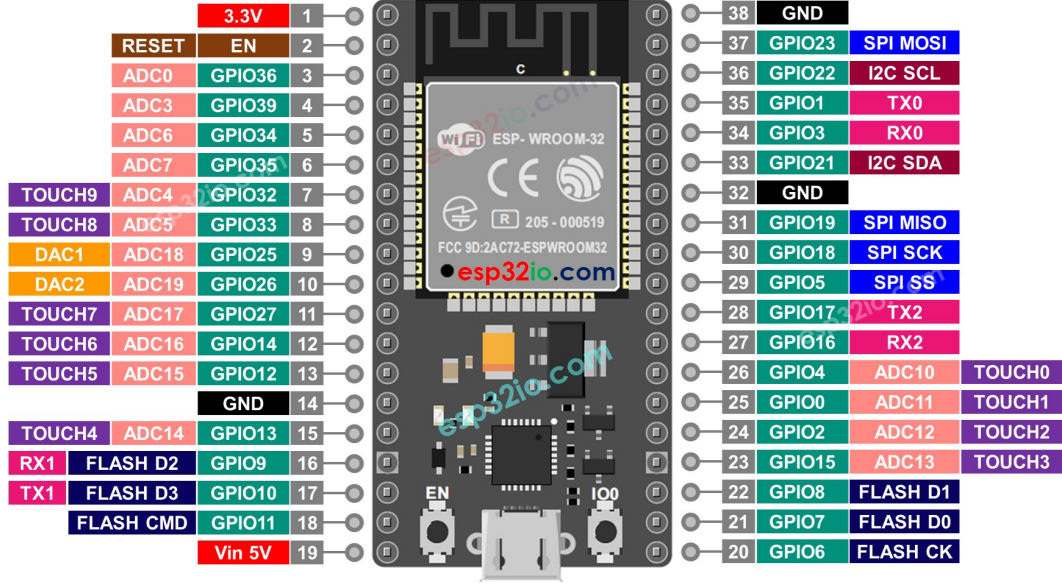


Figure 2.3: ESP32 pin diagram.

## 2.5 Landslide Susceptibility Mapping(LSM)

Landslide Susceptibility Mapping (LSM) is a technique used to identify areas prone to landslides based on various environmental and geological factors. It involves analyzing terrain features such as slope, soil type, rainfall, vegetation cover, and past landslide occurrences using GIS and remote sensing. In our project, LSM helps assess high-risk zones by integrating real-time sensor data and satellite imagery, enabling early warnings and better disaster preparedness.

## 2.6 Supervised Learning for Classification

Supervised learning is a machine learning approach where a model is trained on labeled data to classify or predict outcomes. In our project, it can be used to classify areas into high-risk and low-risk zones for landslides based on historical data and extracted features from Satellite dataset and real-time sensor inputs for real-time prediction. Features like slope, soil features, rainfall, past landslide occurrences, and geography can be used to train models such as Logistic Regression, Decision Trees, Random Forest, or Neural

Networks and create Susceptibility maps. And in Case of Sensor data obtained from IoT system. Once trained, the model can analyze new data from IoT sensors to provide real-time risk assessments and early warnings, improving disaster preparedness and road safety.

## **CHAPTER 3**

### **LITERATURE REVIEW**

#### **3.1 Introduction**

Landslides pose a significant threat to life and infrastructure, particularly in mountainous regions like Nepal. The combination of steep terrain, intense monsoon rainfall, and fragile geology makes the country highly susceptible to frequent landslides. The integration of Geographic Information Systems (GIS), Remote Sensing, Machine Learning (ML), and Internet of Things (IoT) technologies has emerged as a promising approach for effective landslide susceptibility mapping and early warning systems. This chapter reviews existing research on these technologies and their application in landslide monitoring, highlighting the gaps and opportunities for improvement.

#### **3.2 Satellite Imagery and Machine Learning for Landslide Susceptibility Mapping**

Satellite imagery plays a crucial role in landslide susceptibility mapping by providing large-scale geospatial data for analysis. Recent advancements in remote sensing have enabled the use of Sentinel-1 and InSAR (Interferometric Synthetic Aperture Radar) data for accurate landslide prediction. Wang et al. [2] demonstrated how integrating Sentinel-1 and InSAR data improves mapping accuracy in China, proving the effectiveness of remote sensing techniques in landslide detection.

A similar approach has been applied in Nepal, where high-resolution geospatial data is limited. Shrestha and Thapa [3] utilized freely available datasets such as Sentinel-2 and Landsat imagery to develop susceptibility maps. Their study underscores the importance of open-source satellite data in resource-constrained regions.

Machine learning algorithms have further enhanced landslide susceptibility mapping by integrating multiple environmental factors, such as slope, soil type, precipitation, and land use. Studies along the Karakoram Highway [4] found that ML models, particularly Random Forest and Support Vector Machines, significantly outperformed traditional statistical methods. The inclusion of advanced topographic indices like Topographic

Wetness Index (TWI), Topographic Position Index (TPI), Stream Power Index (SPI), and Terrain Ruggedness Index (TRI), as explored in the MDPI study [5], has further improved susceptibility modeling. By extracting these features from various satellite datasets and structuring them into CSV files, researchers can train ML models more effectively, generating probability maps that are later integrated into QGIS for visualization.

### **3.3 IoT-Based Early Warning Systems for Landslide Detection**

IoT-based monitoring systems have gained traction as effective tools for real-time landslide detection. Sensors measuring soil moisture, rainfall intensity, ground vibrations, and slope movement can provide early warnings before a landslide occurs. Research by PSiborg [6] and E3S Conferences [7] demonstrated the use of IoT sensors to continuously monitor environmental conditions and trigger alerts when predefined thresholds are exceeded.

Pawar et al. [8] introduced an IoT-based system that classifies regions into SAFE, MIDDLE, and DANGER zones based on real-time sensor data. Their approach aligns with the proposed IoT component of this project, where collected sensor data is integrated with ML models for dynamic risk assessment. However, existing IoT solutions often focus on localized monitoring and lack integration with broader satellite-based susceptibility mapping. The IIIS project [9] has explored IoT and satellite data fusion, which enhances both monitoring and prediction capabilities, paving the way for comprehensive early warning systems.

### **3.4 Gaps in Existing Research**

Despite advancements in landslide susceptibility mapping and IoT-based monitoring, several challenges remain unaddressed. One major limitation is the lack of integration between real-time IoT monitoring and large-scale satellite-based mapping. Current IoT implementations provide localized data but lack a spatially comprehensive view, whereas satellite-based mapping offers broad coverage but lacks real-time responsiveness.

Another gap is the limited adoption of topographic indices like TWI, TPI, SPI, and TRI in susceptibility models. The MDPI study [5] demonstrated their importance, but their full potential remains underutilized in Nepal. Additionally, most existing research focuses on post-landslide analysis rather than predictive modeling. By leveraging ML-based probability maps along with real-time sensor data, a more proactive approach can be achieved.

### **3.5 Conclusion**

The review highlights the significant progress made in landslide susceptibility mapping using remote sensing and ML, as well as real-time monitoring through IoT. However, integrating these technologies into a unified system remains a critical research gap. By combining ML-based susceptibility mapping with IoT-based real-time monitoring, a more robust and scalable landslide early warning system can be developed. This approach will enhance disaster preparedness in Nepal, ensuring timely alerts and improved risk mitigation strategies.

## **CHAPTER 4**

### **METHODOLOGY**

#### **4.1 Methodology**

Our approach involved two key phases: Landslide Susceptibility Mapping (LSM) and IoT-based real-time monitoring. First, we generated LSM using satellite datasets and historical landslide data from the NASA Global Landslide Catalog. We processed terrain features such as slope, aspect, and elevation, extracting geospatial indices like TPI, TWI, TRI, and SPI from Digital Elevation Models (DEM). After training multiple machine learning models, XGBoost performed best for identifying high-risk areas along the Mugling-Narayanghat highway.

Based on the LSM results, we selected critical zones for real-time IoT monitoring. Our system integrates MPU6050, Soil Moisture Sensor, BMP180, and DHT11, along with precipitation data from an API. Sensor data is transmitted via WebSocket to a cloud-based system, where a trained model predicts landslide risk. Threshold-based classification is used for MPU6050 data, and SMS alerts are triggered when risk exceeds a set threshold.

Although real-time sensor data is not currently aligned with satellite imagery, future improvements include GIS-based visualization. To ensure reliability, we designed a simulation environment mimicking real-world conditions for data collection and model validation.

#### **4.2 Requirement Analysis**

We first conducted a thorough analysis to identify the necessary components for our landslide monitoring system. The project was divided into two main phases: Landslide Susceptibility Mapping (LSM) and IoT-Based Real-Time Monitoring. For LSM, we utilized satellite datasets, including DEM-derived features like slope, aspect, and elevation, precipitation, soil features like bulk density, clay density, sand content etc to generate risk maps for the Mugling-Narayanghat highway. Historical landslide data from the NASA Global Landslide Catalog was used to train machine learning models,

with XGBoost emerging as the most effective. Once high-susceptibility zones were identified, we designed an IoT-based system to monitor ground conditions in real time. The IoT system comprises an ESP32 microcontroller interfacing with sensors such as MPU6050 for ground movement detection, a soil moisture sensor for water content measurement, BMP180 for pressure and altitude tracking, and DHT11 for temperature and humidity monitoring and precipitation value from API. A cloud-based processing system was implemented to analyze sensor data transmitted via Wi-Fi using WebSockets, with alerts being triggered when the risk threshold exceeded 0.8. The entire system is designed to provide early warnings via SMS to mitigate potential disasters, ensuring both accuracy and real-world applicability.

### 4.3 Development

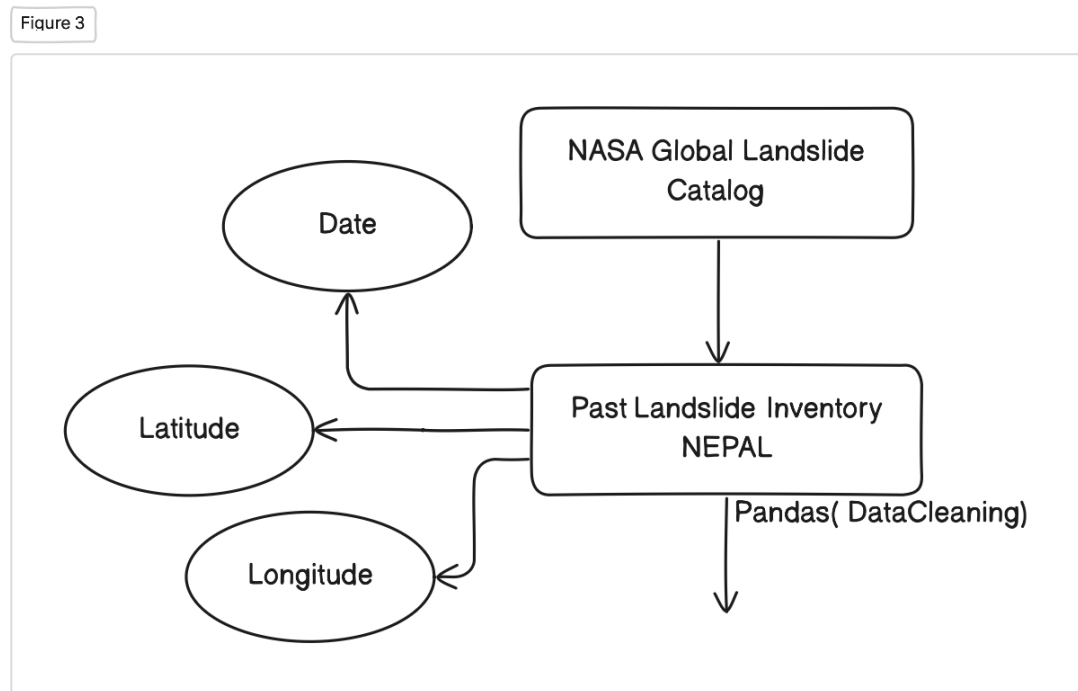


Figure 4.1: historical Landslide data Gathering

To identify high-risk landslide areas along the Mugling-Narayanghat highway, we developed a Landslide Susceptibility Map (LSM) using satellite data, terrain features, and machine learning. We first gathered historical landslide data from the NASA Global Landslide Catalog and selected landslide locations. To create a balanced dataset, we generated non-landslide points at least 1 km away from known landslides, ensuring



reliable classification.

For each point, we extracted topographic, geological, hydrological, and environmental features, including slope, aspect, elevation, distance to roads/rivers/faults, soil properties (sand, silt, clay content, pH, organic carbon), and precipitation trends. Additionally, land use land cover (LULC) and NDVI were used to assess vegetation impact. The dataset was structured as a binary classification problem, where 1 represented landslide-prone areas and 0 represented stable regions.

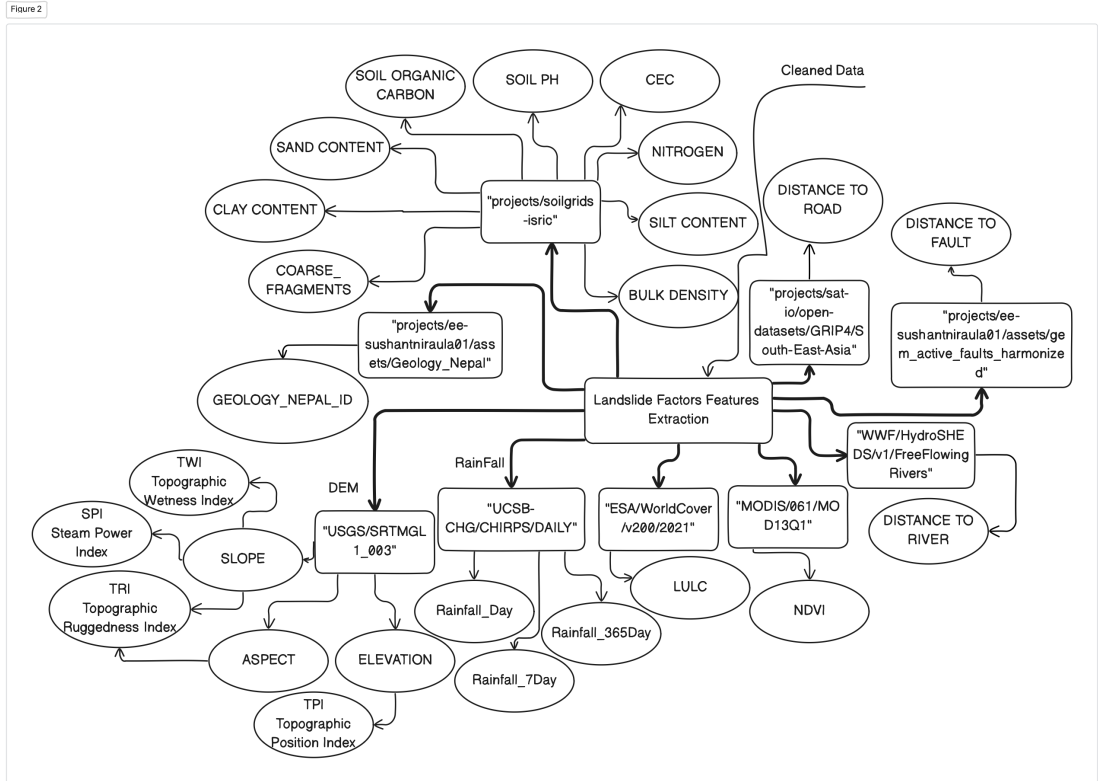


Figure 4.2: Feature Extraction of Landslide datapoints

To train the model, we tested multiple algorithms, including Logistic Regression, Random Forest, and XGBoost, with XGBoost delivering the best performance after hyperparameter tuning and cross-validation. The trained model was then used to predict landslide susceptibility across a high-resolution spatial grid. We generated 40,000 points at 100m resolution across the study area and extracted the same features for these locations. The model provided probability scores for each point, indicating landslide risk levels.

The final susceptibility map was prepared using QGIS, where predicted probabilities were visualized with a color gradient, highlighting high-risk zones. This LSM served

as the foundation for selecting locations for IoT-based real-time monitoring, ensuring a data-driven approach to landslide risk assessment and mitigation.

After identifying high-risk zones through Landslide Susceptibility Mapping (LSM), we developed an IoT-based real-time landslide prediction system to monitor ground conditions continuously. The system was designed to collect real-time environmental and geotechnical data, process it in a cloud-based system, and trigger alerts when landslide risk exceeded a defined threshold.

The IoT system consists of multiple sensors, including MPU6050 (for ground movement detection), Soil Moisture Sensor, BMP180 (for atmospheric pressure), and DHT11 (for temperature and humidity). Since a rain gauge was unavailable, we integrated precipitation data from an API to include rainfall trends. These sensors were powered through a stable power cable connection and sampled data every 15 seconds. WebSocket communication was used to transmit sensor readings to a cloud-based processing system for real-time analysis.

For risk prediction, we trained a machine learning model using past landslide events and sensor readings. The MPU6050 sensor data was processed using statistical methods, where the mean acceleration values were computed, and a threshold-based risk classification was set. The trained model analyzed incoming sensor data and generated a risk probability score. If the probability exceeded 0.8, an SMS alert was triggered to notify authorities and relevant stakeholders.

To ensure reliability, we designed a simulation environment mimicking real-world conditions, allowing us to collect more representative data before deployment. Although real-time sensor data is not currently aligned with satellite imagery, future improvements will include GIS-based visualization to integrate both sources. The system is scalable, allowing deployment in any high-risk region by first generating an LSM map and then implementing IoT-based monitoring for real-time risk assessment.

#### **4.4 System Architecture**

The system architecture integrates satellite-based Landslide Susceptibility Mapping (LSM) with an IoT-powered real-time monitoring system, ensuring both long-term risk

assessment and immediate alerts. The design follows a layered approach, where satellite data informs risk zones, and IoT sensors provide continuous monitoring in high-risk areas.

The data collection layer gathers historical landslide data from the NASA Global Landslide Catalog, along with terrain and environmental features from remote sensing datasets. IoT sensors, including MPU6050, Soil Moisture Sensor, BMP180, and DHT11, collect real-time data, while precipitation data is obtained from an external API.

In the data processing layer, LSM is generated by extracting features like slope, elevation, NDVI, soil properties, and distances to roads, rivers, and faults. A balanced dataset was created by generating non-landslide points at least 1 km from known landslides. After training multiple models, XGBoost performed best and was used to predict landslide probability for 40,000 grid points (100m resolution) along the Mugling-Narayanghat highway.

For real-time monitoring, IoT sensor data is transmitted via WebSocket over WiFi to a cloud-based system. The ML model predicts landslide risk, while MPU6050 motion values are processed statistically using threshold-based classification. When the risk probability exceeds 0.8, an SMS alert is triggered to warn relevant authorities.

The system is scalable, allowing rapid LSM generation for any polygonal area before deploying IoT sensors for real-time monitoring. While GIS-based visualization is not yet integrated, QGIS was used for mapping LSM predictions. Future enhancements will include real-time GIS-based monitoring and improved data fusion techniques.

## **4.5 Testing**

Testing was conducted in two phases: model validation for Landslide Susceptibility Mapping (LSM) and real-time IoT system evaluation.

For LSM testing, we evaluated multiple models, including Logistic Regression, Random Forest, and XGBoost, using Accuracy, Recall, and F1-score. The dataset was balanced by generating non-landslide points, ensuring a fair evaluation. XGBoost performed best after hyperparameter tuning, and its predictions were validated against known landslide occurrences.

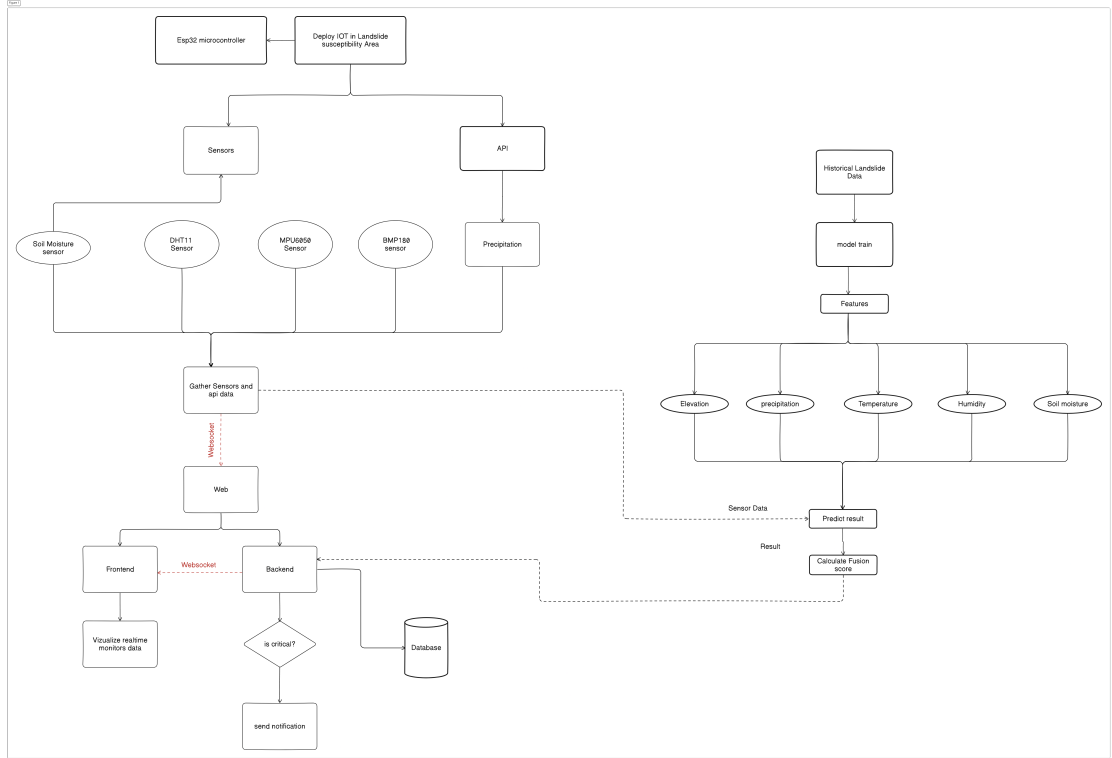


Figure 4.3: IOT Implementation Diagram

For IoT system testing, we got the dataset from the web similar to our sensors data and then we build a model on it. The data was highly imbalanced so we used various Techniques like SMOTE, ADASYN, UNDERSAMPLING, OVERSAMPLING and found that Gradient boost model with SMOTE worked the best and then we designed a simulation environment mimicking real-world conditions to test model on sensor data. The machine learning model was tested on this data, and threshold-based risk classification was applied for MPU6050 motion readings. WebSocket communication, cloud-based data processing, and SMS alerts were verified to ensure smooth real-time operation.

While the system is not yet deployed in the Mugling-Narayanghat highway, testing in a controlled environment ensures reliability and accuracy, with plans for further validation in the next monsoon season.

## 4.6 Technology Stack and Environment Setup

### 4.6.1 Backend

- **Language:** Node.js

- **Database:** Postgres
- **SMS Service:** Infobip

#### 4.6.2 Frontend

- **Framework:** NextJs

### 4.7 Detailed Implementation Steps

#### 4.7.1 Backend Implementation

- **Data Collection:** Gather environmental data (rainfall, soil moisture, etc.).
- **Data Preprocessing:** Clean and normalize the data.
- **Model Training:** Train a machine learning model for landslide prediction.

#### 4.7.2 Frontend Implementation

- **User Interface:** Developed as a Single-Page Application (SPA) using Nextjs.
- **Data Visualization:** Display predictions and data using charts.
- **Protocol:** Used Websocket for seamless Data Transfer.
- **Geospatial Data:** Show prediction results on interactive maps.

### 4.8 Diagrams

#### 4.8.1 Use Case Diagram

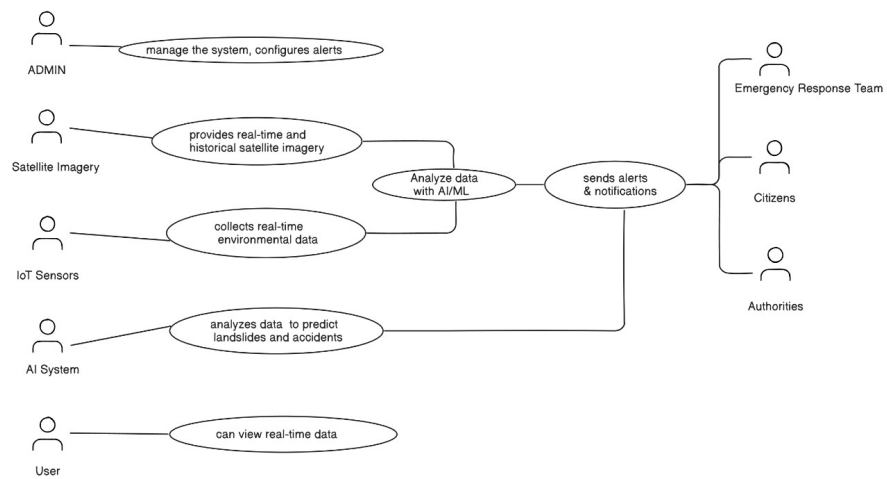


Figure 4.4: Use Case Diagram

## 4.8.2 ER Diagram

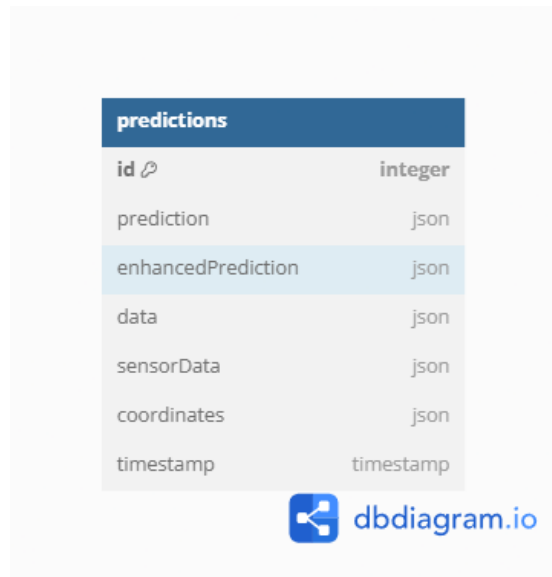


Figure 4.5: ER Diagram

## CHAPTER 5

### RESULTS

#### 5.1 Landslide Susceptibility Mapping (LSM)

The LSM for the Muglin-Narayanghat highway was generated using satellite imagery, past landslide inventory, and geographical data. A dataset of past landslide locations was combined with non-landslide points (selected at least 1 km away) to train and evaluate models. XGBoost performed best and was used to predict landslide susceptibility.

A total of 40,000 points were processed to create the final LSM, classifying areas into Low (less than 20%), Moderate (20%-50%), and High (Greater than 50%) risk zones. The results were visualized in QGIS, helping to identify high-risk areas for further monitoring and mitigation.

##### 5.1.1 Prediction Output and Risk Assessment

The prediction model assigns a risk level to various geographical regions based on input data such as rainfall, soil moisture, slope angle, and historical landslide occurrences. The model categorizes risk levels into three distinct classes:

- **Low Risk:** Less than 40% chance of a landslide occurrence.
- **Moderate Risk:** Between 40% and 80% chance of a landslide.
- **High Risk:** More than 80% chance of a landslide.

During initial testing with real-world data, the model successfully identified several high-risk areas, such as regions characterized by steep slopes and recent heavy rainfall.

This risk assessment approach enhances the system's ability to provide actionable insights, enabling local authorities and residents to take timely precautions to mitigate the impact of landslides.



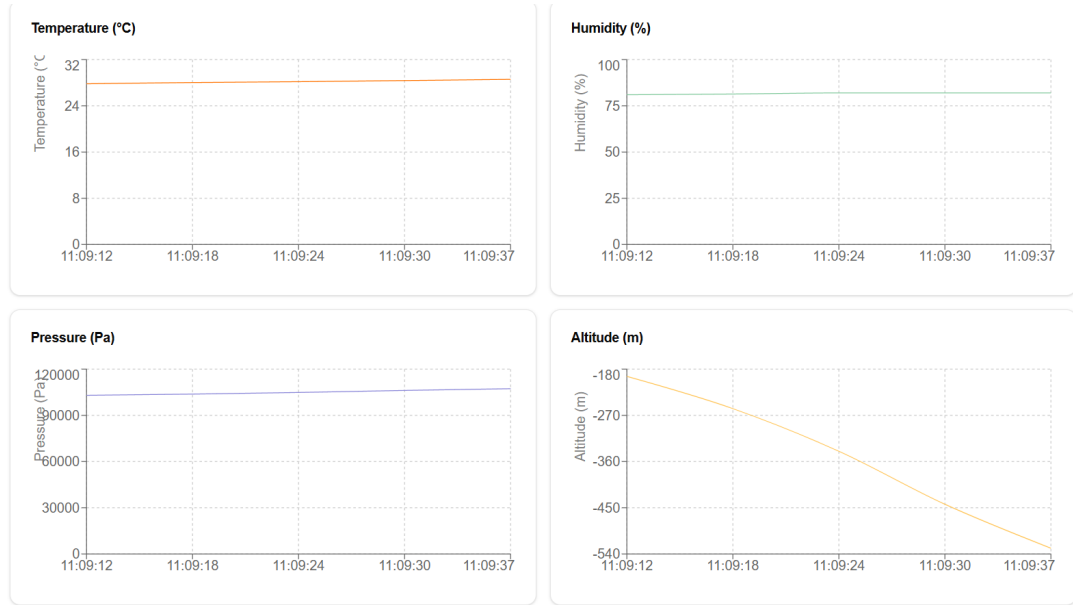


Figure 5.1: Temperature Dashboard

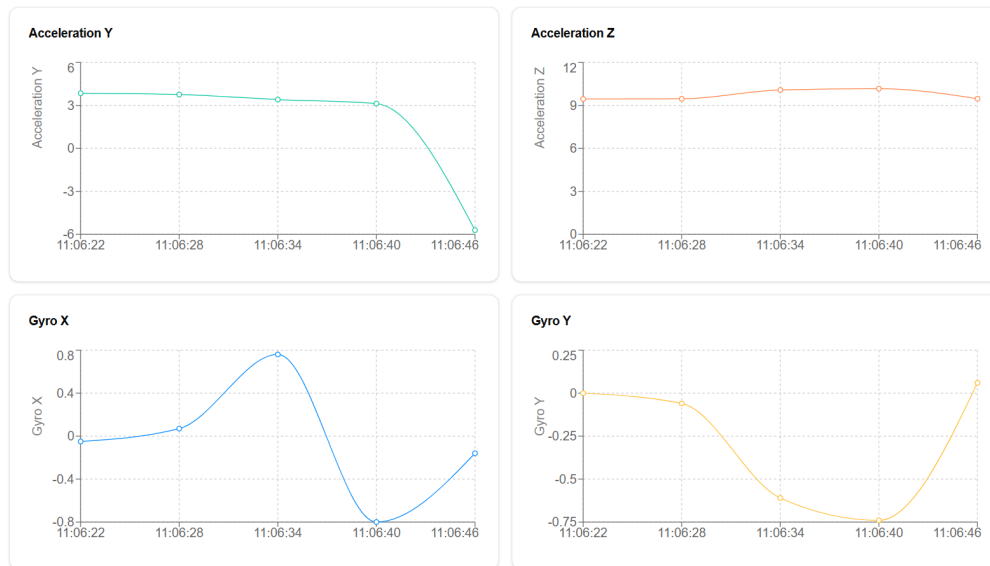


Figure 5.2: Dashboard

## 5.2 Real-Time Alerts and User Interface

The early landslide prediction system includes a real-time alert feature that sends notifications to users, authorities in High-Risk zones based on predicted risk levels. These alerts are delivered through a web application, enabling local authorities and residents to take precautionary actions. The user interface is designed for simplicity, displaying predicted landslide risks for various regions, making it easy for users to understand the risk levels in their areas.

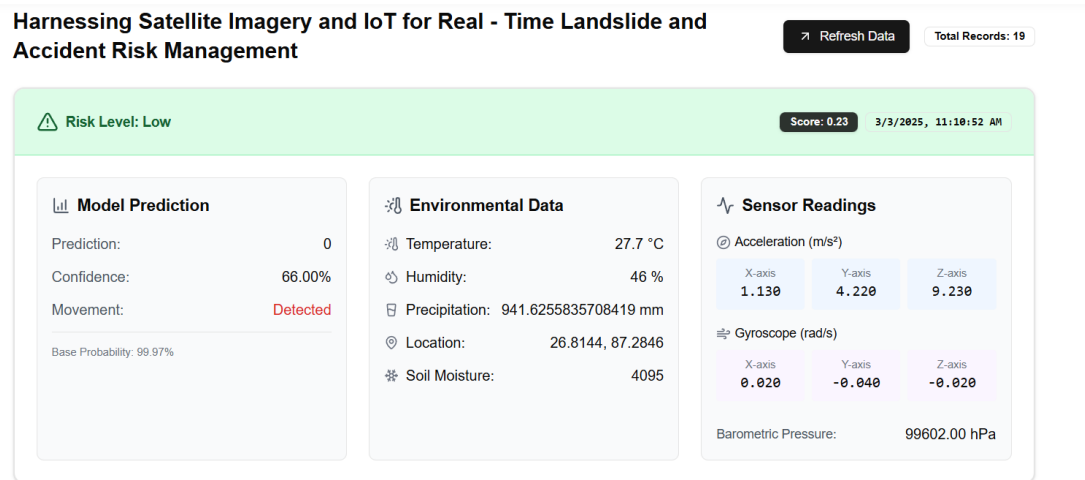


Figure 5.3: Low Risk Measurement

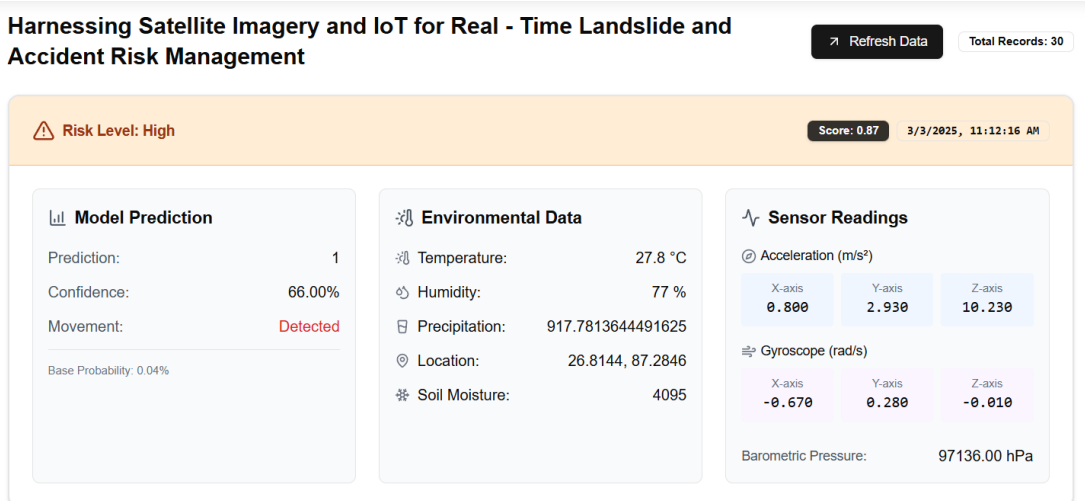


Figure 5.4: High Risk Measurement

## CHAPTER 6

### CONCLUSION

This project successfully integrates Landslide Susceptibility Mapping (LSM) using machine learning and real-time IoT-based monitoring to enhance landslide risk assessment and early warning systems. The combination of satellite-based geospatial analysis and sensor-driven real-time data collection provides a comprehensive approach to landslide prediction and monitoring.

The LSM model was developed using terrain-based features such as Topographic Position Index (TPI), Terrain Ruggedness Index (TRI), and other geomorphological attributes, inspired by methodologies presented in MDPI research studies [5]. These features significantly improved landslide prediction by capturing the influence of slope variations and landform characteristics. The Muglin-Narayanghat highway susceptibility map, generated from 40,000 data points, highlighted high-risk zones, confirming the effectiveness of using machine learning for susceptibility assessment [2].

The IoT-based real-time monitoring system complements the LSM by providing continuous environmental sensing. Inspired by IoT-driven early warning frameworks [7], we deployed MPU6050, soil moisture, BMP180, and DHT11 sensors, integrating WebSocket-based real-time data transmission for cloud processing. This system demonstrated the potential of sensor-based monitoring for immediate landslide risk detection, aligning with studies emphasizing IoT's role in disaster mitigation [8].

By bridging geospatial data analysis with IoT-based real-time sensing, this project presents a scalable landslide risk management framework. Future work includes expanding GIS-based visualization tools, enhancing sensor accuracy with advanced filtering techniques, and refining prediction models with additional environmental variables. This project contributes to disaster risk reduction efforts, reinforcing the importance of integrating machine learning, GIS, and IoT for proactive hazard management [9].

## **6.1 Further Improvements And Challenges**

The project faced key challenges due to the lack of high-resolution data, incomplete past landslide databases, and limited field surveys in Nepal. These gaps affected the accuracy of the Landslide Susceptibility Mapping (LSM) model and restricted validation with real-world data. More comprehensive datasets and improved field surveys are crucial for enhancing model reliability.

With high-resolution geospatial data and better computing resources, higher-resolution LSM maps could be generated, improving risk assessment. The IoT system, currently tested in a controlled environment, requires further experimental simulations before full deployment in high-risk zones.

Future enhancements could integrate cloud-based GIS, deep learning for geospatial analysis, and improved environmental modeling, making the system a robust landslide early warning tool for Nepal.

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